

Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 2



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Recap of Previous Lecture



Topic

Regular language

Topic

Topic

Topics to be Covered



Topic

Regular languages

Topic

Topic

Properties of Regular expressions -
 let r, s, t be Regular expressions

commutative property -

$$r + s = s + r$$

$$a^* + b^* = b^* + a^*$$

$$ab^* + a^* = a^* + ab^*$$

$\gamma.S \neq S.\gamma$ $\rightarrow$ may Δ may not

$$a.b \neq \underline{b.a}$$

$$\underline{a^*} \cdot a = \underline{a} \cdot \underline{a^*} = a^+$$

$$a^* \cdot a = \{\epsilon, a, aa, aaa, \dots\} \cdot a_{1,2,3,4,\dots}$$

$$= \{a, aa, aaa, \dots\} = a^+ \rightarrow \text{positive closure}$$

Associative property:-

$$r + (s + t) = (r + s) + t$$

$$a^* + (b + c^*) = (a^* + b) + c^*$$

$$r(st) = (rs)t$$

$$a^*(bc^*) = (a^*b)c^*$$

Distributive property:-

$$r(s+t) = rs + rt$$

$$a^*(bc+d) = a^*bc + a^*d$$

$$r + (s.t) \neq (r+s)(r+t)$$

$$a+bc \neq (a+b)(a+c)$$

$$L = \{\underline{a}, \underline{bc}\}$$

$$a+bc \neq \overbrace{(a+b)(a+c)}^{L = \{aa, ac, ba, bc\}}$$

Identity:

$$\gamma + \phi = \gamma = \overbrace{\phi + \gamma}$$

$\downarrow$ $\downarrow$
 $\{\}$ $\{\}$ ✓

$$\gamma \cdot \epsilon = \gamma = \epsilon \cdot \gamma = \epsilon \cdot \gamma \cdot \epsilon$$

$\gamma \cdot \phi = \phi$

$$a^{*} = (a^0, a^1, a^2, a^3, \dots)$$

$\downarrow$
 ϵ

$$\gamma + \epsilon \neq \gamma \rightarrow \text{may } \delta \text{ may not}$$

$$\underbrace{(a^{*}) + (\epsilon)}_{=} = a^{*}$$

$$\underbrace{(a^{*}) + (\epsilon)}_{=} = a^{*}$$

$\left\{ a, aa, aaa, \dots \right\}$

$$\begin{array}{c} \epsilon + \phi = \\ \downarrow \downarrow \\ \{\epsilon\} \cup \{\} = \{\epsilon\} \end{array}$$

$$L\{\gamma_1 + \gamma_2\} = L(\gamma_1) \cup L(\gamma_2)$$

$$L(\gamma_1 \cdot \gamma_2) = L(\gamma_1) \cdot L(\gamma_2)$$

$$\epsilon \cdot \phi = \{\epsilon\} \cdot \{\} = \{\} = \phi$$

$$\phi^* = \epsilon$$

$$\phi^+ = \phi$$

$$(a^* \cdot \phi)^* = (\phi)^* = \epsilon$$

$$\gamma + \gamma = \gamma \quad \phi \cdot \phi = \phi$$

$\gamma \cdot \gamma \neq \gamma \rightarrow$ may or may not

$$a \cdot a \neq a \downarrow L = \{aa\} \neq L = \{a\}$$

$$\underbrace{(a+b)^*}_{\checkmark} \underbrace{(a+b)^*}_{\otimes} = (a+b)^*$$

$$\underline{a}^* \cdot \underline{a}^* = \underline{a}^*$$

$$\underline{a}^* \cdot \{ \epsilon, a, aa, aaa, \dots \}$$

$$\{ \underline{a}^*, \underline{a}^*a, \underline{a}^*aa, \dots \}$$

$$\gamma^* + \gamma^* = \gamma^*$$

$$a^* + \underbrace{(a^*)}_{\text{circled}} = a^*$$

$$(a+b)^* + \underline{(a+b)^*} = (a+b)^*$$

$$\gamma^* \cdot \gamma^* = \gamma^*$$

$$(\gamma^*)^* = \gamma^*$$

$$(a^*)^* = \{\epsilon, a, aa, aaa, \dots\}^* = a^*$$

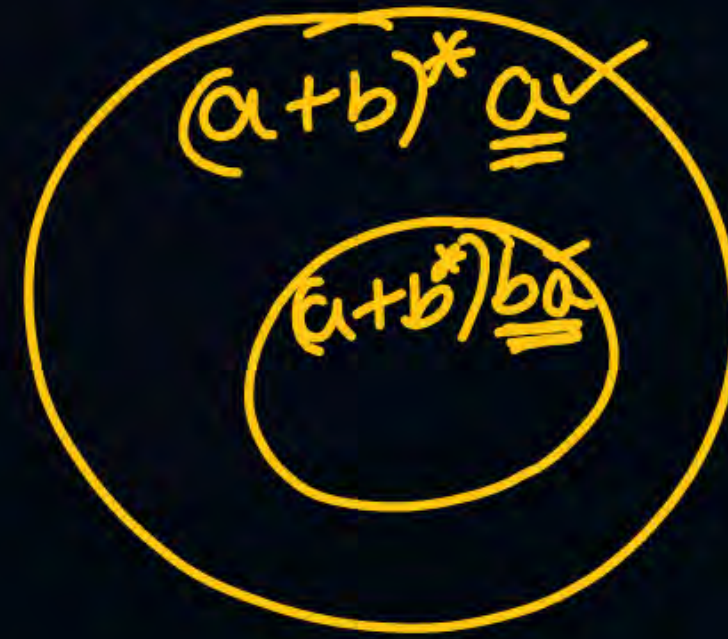
$$\begin{aligned}
 (\gamma^*)^+ &= \gamma^* \\
 (\gamma^+)^* &= \gamma^* \\
 (\gamma^+)^+ &= \{\gamma, \gamma\gamma, \gamma\gamma\gamma, \dots\} \\
 &= \gamma^+
 \end{aligned}$$

$$\begin{aligned}
 \gamma^* \cdot \gamma^+ &= \gamma^* \cdot \{ \gamma, \gamma\gamma, \gamma\gamma\gamma, \gamma\gamma\gamma\gamma, \dots \} \\
 &= \{ \gamma^* \gamma, \gamma^* \gamma\gamma, \gamma^* \gamma\gamma\gamma, \dots \} \\
 &= \gamma^+ \quad \quad \quad = \gamma^+
 \end{aligned}$$

$$\gamma^+ \cdot \gamma^* = \gamma^+$$

$$\begin{aligned}
 \gamma^+ \cdot \gamma^+ &= \gamma^+ \{ \gamma, \underbrace{\gamma\gamma, \gamma\gamma\gamma, \dots}_{\text{odd}} \} \\
 &= \underbrace{\gamma^+ \gamma}_{\text{even}} + \underbrace{\gamma^+ \gamma\gamma + \gamma^+ \gamma\gamma\gamma + \dots}_{\text{odd}} \\
 &= \gamma^+ \gamma
 \end{aligned}$$

$$\frac{\gamma^+ \cdot \gamma^+}{\gamma\gamma} = \frac{\gamma\gamma^+}{\gamma\gamma} = \frac{\gamma^+ \gamma}{\gamma\gamma} = \gamma \overset{\circ}{\underset{\gamma\gamma}{*}} \gamma = \underset{\gamma\gamma}{\gamma\gamma} \gamma \overset{\circ}{\underset{*}{\circ}} = \underset{\gamma\gamma}{\gamma\gamma^*} \gamma$$



If $L(\underline{S}) \subseteq L(Y)$ then $\underline{Y+S} = \underline{Y}$

$$\underline{(a^*)} + \underline{aa^*} = a^*$$

$$\{\underline{\epsilon}, a, aa, \dots\} \cup \{a, aa, aaa, \dots\}$$

$$a^+ + \underline{aa} = a^+ \checkmark$$

$$(a+b)^*(a+ba) =$$

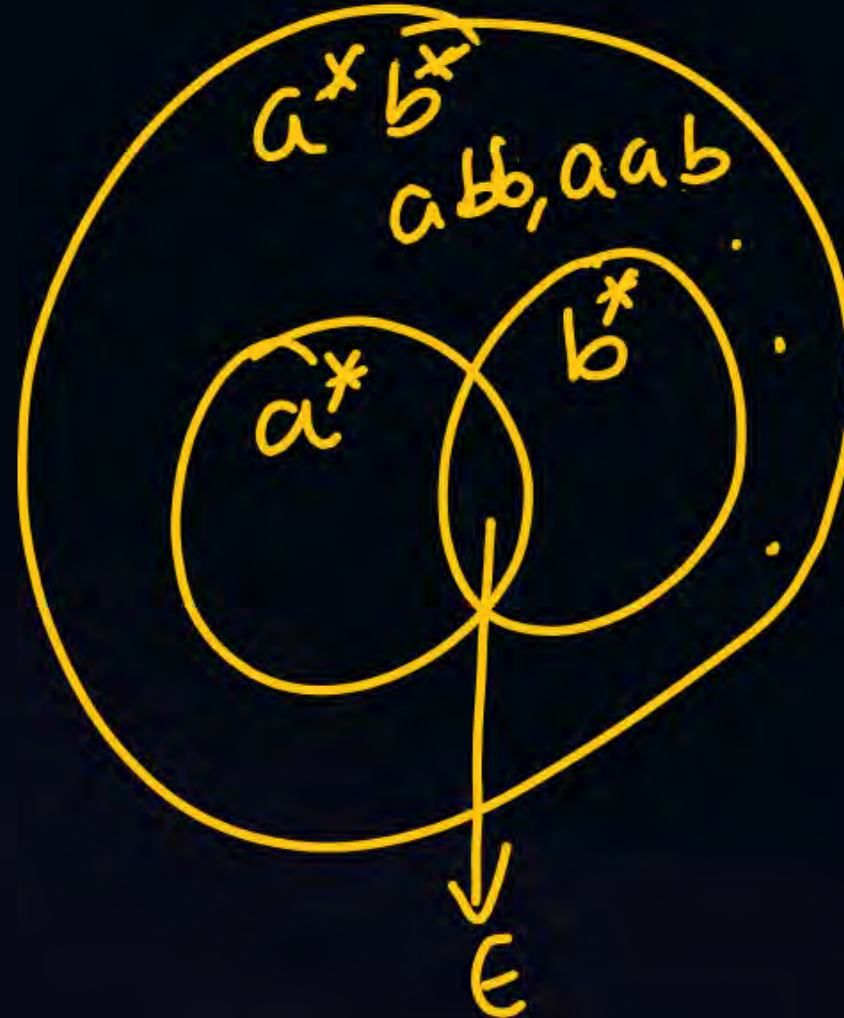
$$(a+b)^*\underline{a} + (a+b)^*\underline{ba} = (a+b)^*a$$

$$\underline{ba} \checkmark$$

$$a^* \overset{\circ}{b^*} + \underline{a^*} =$$

$$\checkmark a^* \subseteq \underline{a^* b^*}$$

$$\checkmark b^* \subseteq \underline{a^* b^*}$$



$$a^* = \{\epsilon, a, aa, aaa, \dots\}$$

$$b^* = \{\epsilon, b, bb, \dots\}$$

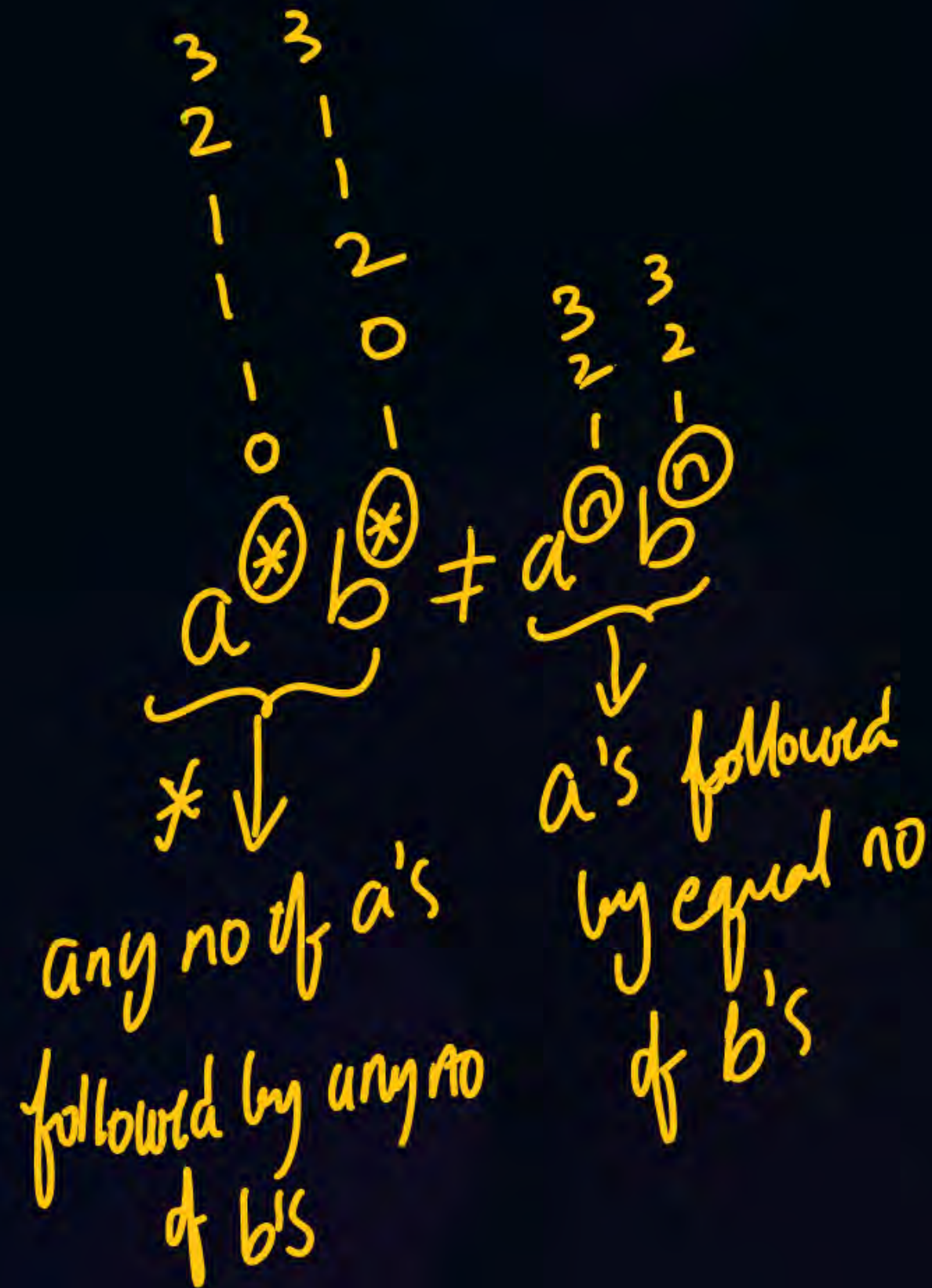
$$(a+b)^* (a^*) = (\underline{a+b})^* \rightarrow \text{universal}$$

$$(a+b)^* (a^* b^*) = (\underline{a+b})^* \rightarrow$$

$$\underline{a^*} b^* \neq a^*$$

$$\underline{a^*} b^* = a^* \cup b^* \cup a^* b^*$$

$$= \{ \epsilon, a, aa, aaa, \dots \} \cup \{ \epsilon, b, bb, bbb, \dots \} \cup \{ ab, aab, abb, \dots \}$$



$$a^*b^* = a^m b^n / m \geq 0, n \geq 0$$

$$\epsilon + \underline{Y} \underline{Y}^* = \underbrace{\epsilon + \underline{Y}^+}_{\underline{Y}^*} = \underline{Y}^* = \underline{Y}^* \underline{Y} + \epsilon = \underline{Y} \underline{Y}^* + \epsilon$$

$$\epsilon + \frac{(a^* b b^*)}{\underline{Y}} \frac{(\underline{a^* b b^*})}{\underline{Y}^*} = (a^* \underline{b b^*})^* = (a^* b^+)^*$$

$$\epsilon + \underline{Y} \underline{Y}^* = \underline{Y}^*$$

$$\begin{aligned}
 & \cancel{\neq} \cancel{\gamma^* + S^*} \\
 & (\gamma + S)^* = (\gamma^{\circ} \otimes^1 S^{\circ})^* = (\gamma + S^{\circ})^* = (\gamma^{\circ} \otimes^1 S)^* = \underline{(\gamma^* S^*)^*} = \underline{(S^* \gamma^*)^*} \\
 & \quad (a+b) \\
 & \quad \underline{\gamma \gamma S S} \\
 & \quad \underbrace{(\gamma^{\circ} \otimes^1 S^{\circ}) (\gamma^{\circ} \otimes^1 S^{\circ})}_{(S^* \gamma^*)^{\circ 4}} \quad \overline{(\gamma^{\circ} \otimes^1 S^{\circ}) (\gamma^{\circ} \otimes^1 S^{\circ})} \quad (\gamma^{\circ} \otimes^1 S^{\circ}) (\gamma^{\circ} \otimes^1 S^{\circ}) \\
 & \quad (\gamma^{\circ} \otimes^1 S^{\circ}) (\gamma^{\circ} \otimes^1 S^{\circ}) \quad (\gamma^{\circ} \otimes^1 S^{\circ}) (\gamma^{\circ} \otimes^1 S^{\circ}) \quad (\gamma^{\circ} \otimes^1 S^{\circ}) (\gamma^{\circ} \otimes^1 S^{\circ}) \quad (\gamma^{\circ} \otimes^1 S^{\circ}) (\gamma^{\circ} \otimes^1 S^{\circ})
 \end{aligned}$$

$$(a+b)^* \neq \underbrace{a^*b^*}_{ba}$$

Diagram illustrating the difference between $(a+b)^*$ and a^*b^* . The expression $(a+b)^*$ is shown with a bracket underneath it, and a circle labeled ba is connected to the bracket by a curved arrow, indicating that $(a+b)^*$ includes the sequence ba . The expression a^*b^* is shown with a bracket underneath it, and a circle labeled ba is connected to the bracket by a curved arrow, indicating that a^*b^* does not include the sequence ba .

- 1) $(a^*b^*)^* \text{ --- } (a+b)^*$
- 2) $(b^*a^*)^* \text{ --- } (a+b)^*$
- 3) $(a^{\circ*} + b^{\circ*})^* \text{ --- } (a+b)^*$
- 4) $(a^{\circ*} + ba^{\circ*})^* \text{ --- } (a+b)^*$
- 5) $(a^{\circ*}b + b^{\circ*}b)^* \neq (a+b)^*$

$$(\gamma + S)^* = (\gamma^{\oplus} + S^{\oplus})^* = (\gamma^{\oplus} + S)^* = (\gamma + S^{\oplus})^*$$

$$(a+b)^* \underbrace{aa^*} \rightarrow ? \underbrace{(a+b)^* a^*} a \rightarrow \underline{(a+b)^* a} \downarrow \underbrace{\text{ending with } a}$$

$$\underline{aa^*} = \underline{a^* a}$$

$$(a+b)^* a^* \neq (a+b)^*$$

$$a^+ b^+ = a^n b^m / n, m \geq 1$$

$$(a+b)^* b b^* \rightarrow (a+b)^* b^* b$$

$\rightarrow (a+b)^* b$
 ending with 'b'

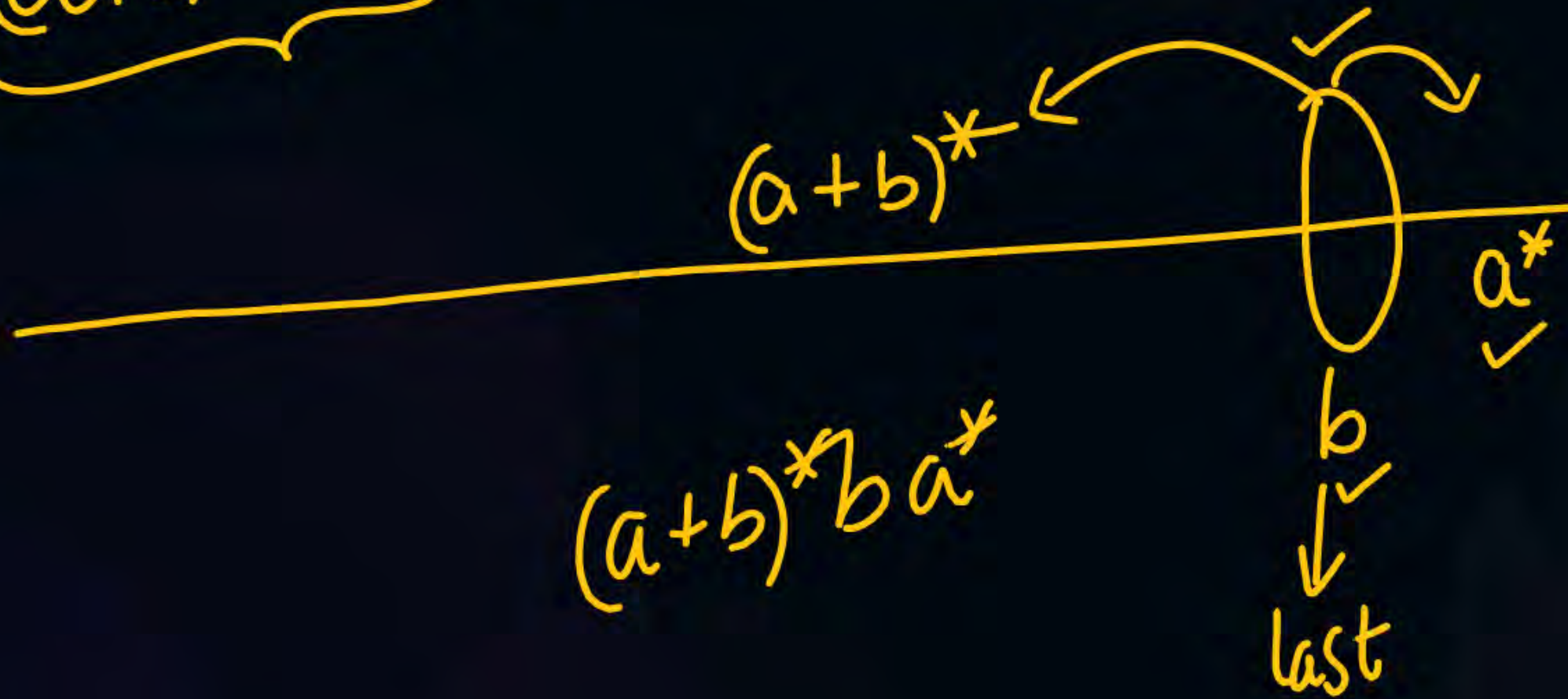
$$(a+b)^* (ba)^*$$

at least one b

$$\underline{ba^*} \stackrel{?}{=} \underline{a^* b} \quad \times$$

$$(a+b)^0 \underline{b} \checkmark$$

$(a+b)^* b a^*$ → at least one b is present



$$(a+b)^* b a^*$$

$$a^* \cup (a+b)^*$$

$$a^* b (a+b)^* = (a+b)^* b a^* \cup (a+b)^* b (a+b)^*$$

Language containing at least 1 b

$$\frac{(a+b)^*}{\text{Random } b} (a+b)^*$$

$$\underbrace{(a+b)^* a^*}_{c \in L} \stackrel{?}{=} \underbrace{(a+b)^* a^*}_{\in}$$

$$(a+b)^* a^* a$$

$$(a+b)^* a^*$$

↓
ending with a

$$(a+b)^* a^*$$

↓
universal language

$$\Sigma = \{a\} \quad L = a^* \rightarrow RE$$



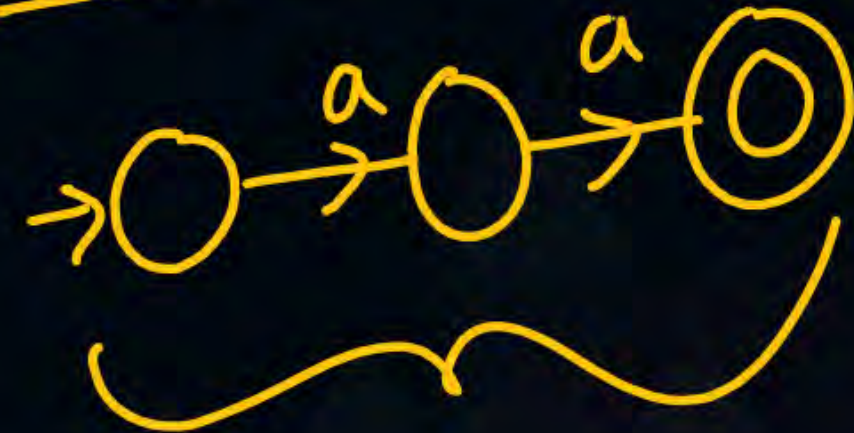
Whenever, initial state is also a final state
in NFA or DFA, then 'E' will be accepted

$\Sigma = \{a\} \quad L = \{\underline{aa}\}$

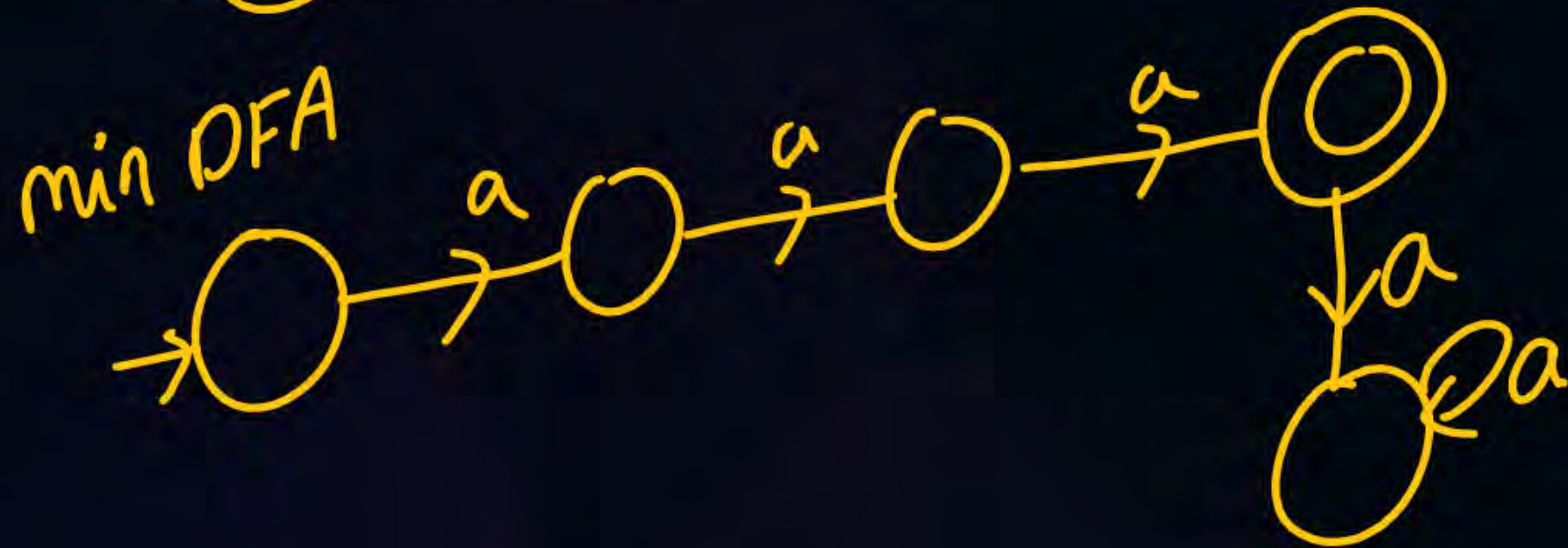
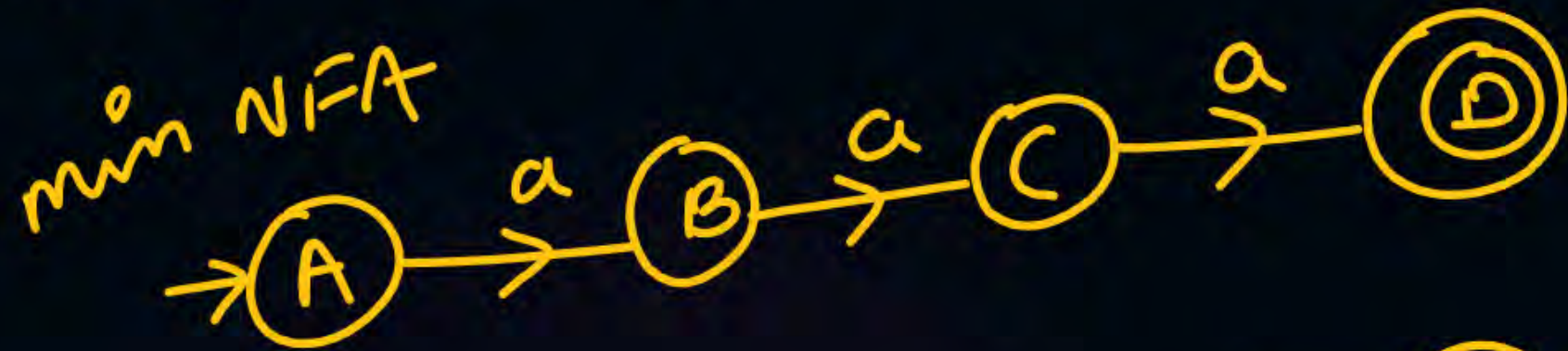
min DFA



min NFA



$\Sigma = \{a\}$ $L = \{aaa\}$

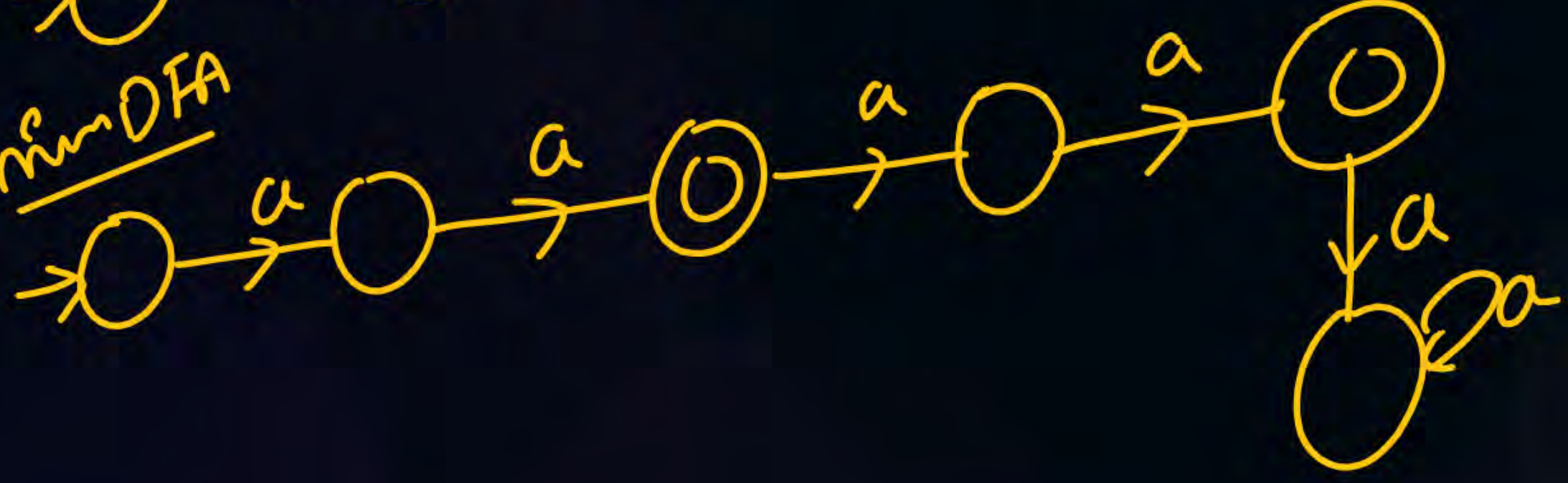


$\Sigma = \{a\}$
 $L = \{aa, aaaa\}$

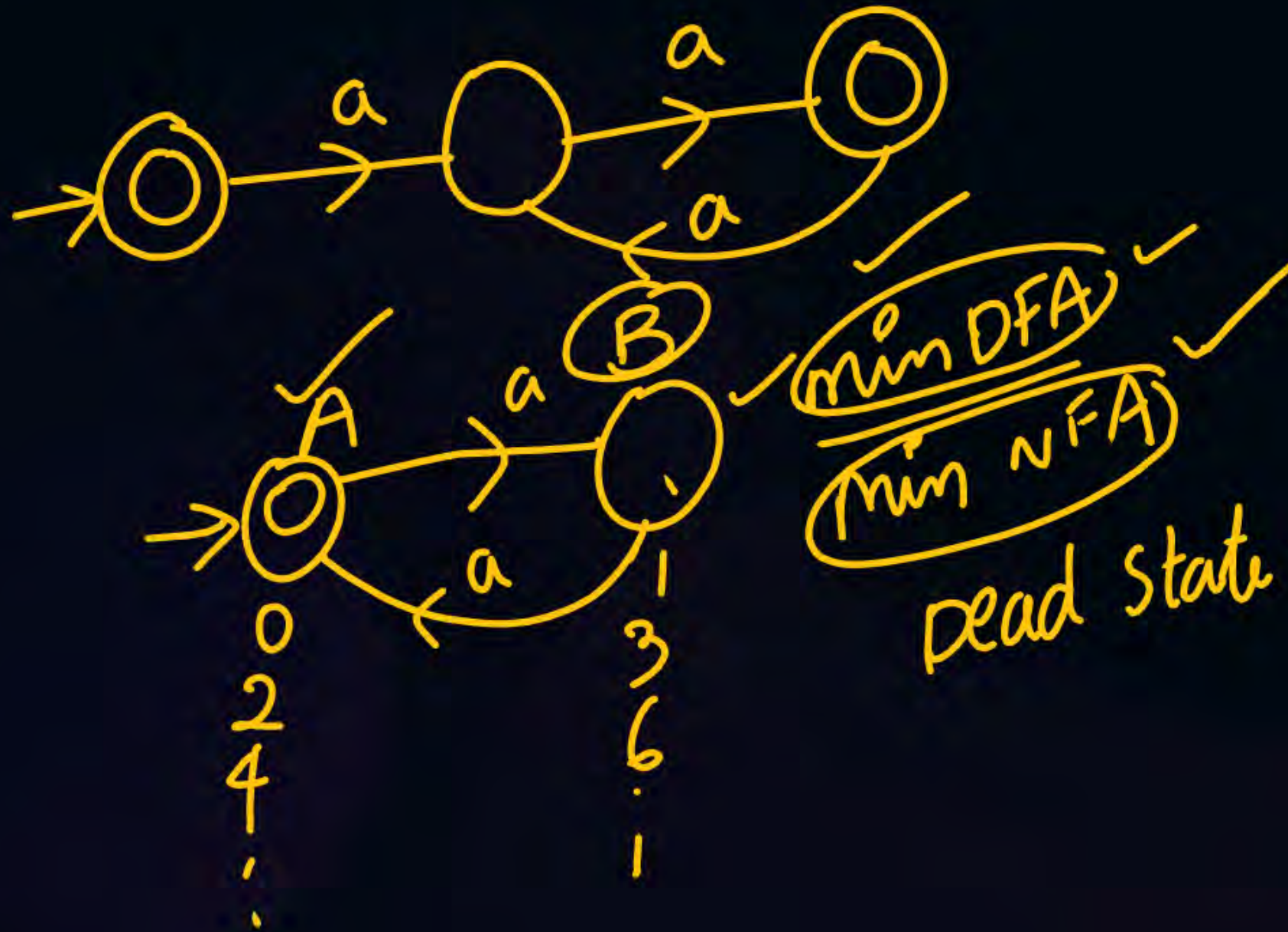
min NFA



min DFA



$\Sigma = \{a\}$ ✓
 $L = \text{even no of } a\text{'s}$ ✓ $(\underline{aa})^* \{ \epsilon, aa, aaaa, aaaaaa, \dots \}$



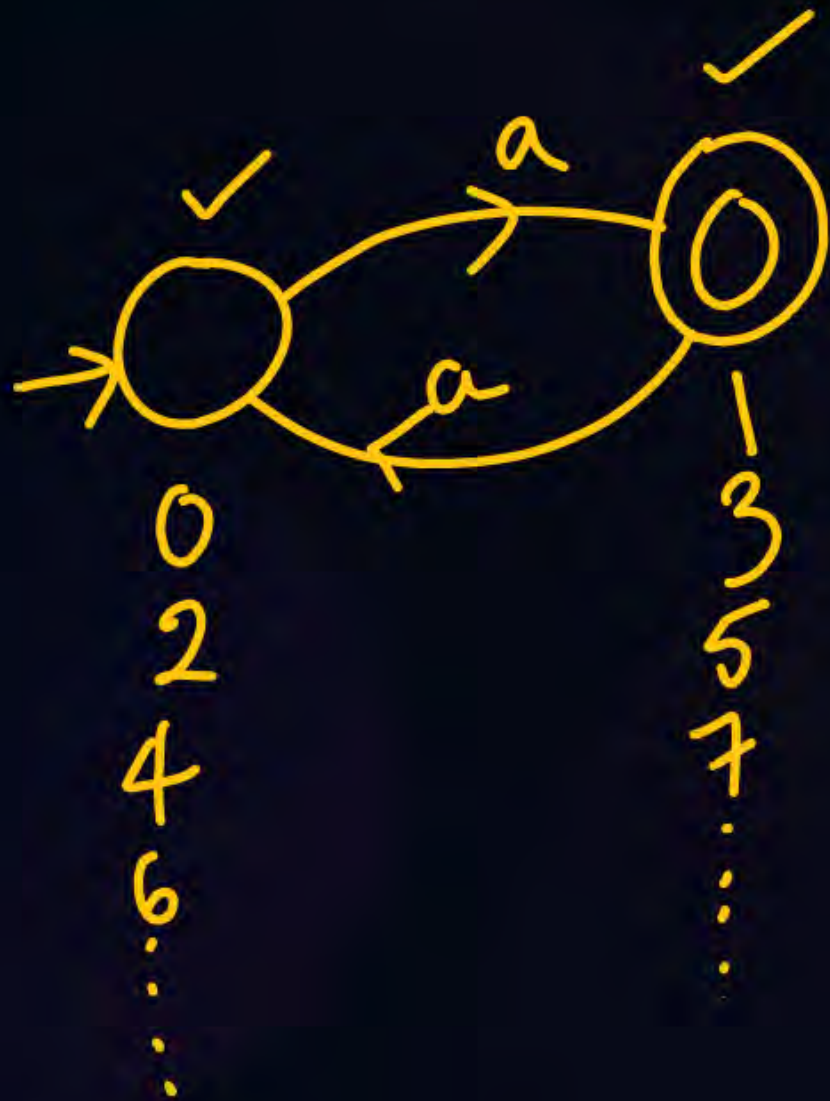
min DFA ✓
 min NFA ✓

dead state → we cannot reach final state from a dead state

$$\Sigma = \{a\}$$

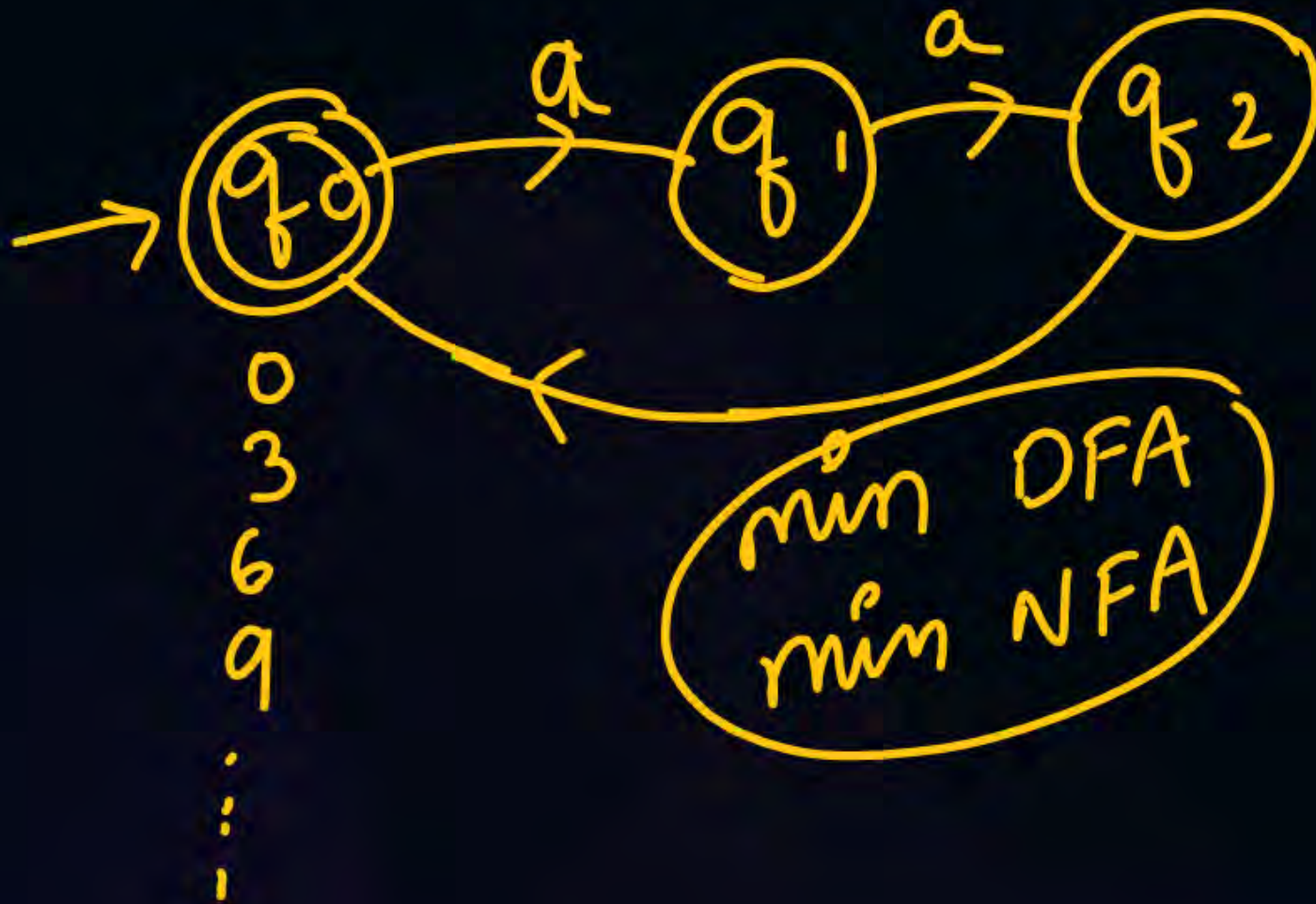
odd no of a's

$$RE = (aa)^*a = a(aa)^* \neq \underbrace{(aaa)^*}_{\text{multiple of 3}}$$



min DFA ✓
&
min NFA ✓

$\Sigma = \{a\}$ $a^{3n} / n \geq 0$ $(aaa)^*$



min DFA
min NFA

$a^{100n} / n \geq 0$

↓
state?

100 ✓

$$\Sigma = \{a\}$$

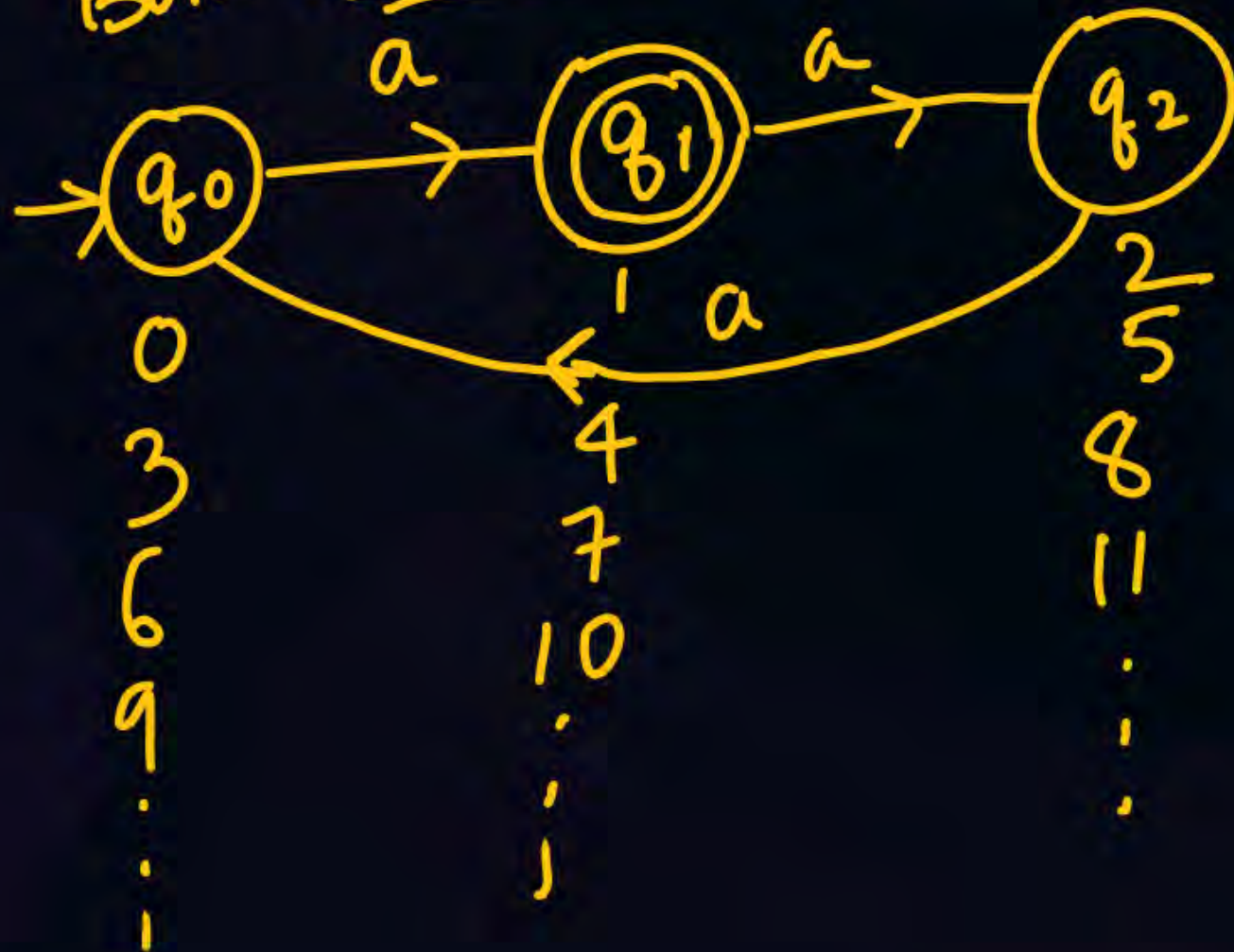
$$L = \text{no of } a\text{'s} \equiv 1 \pmod 3$$

$$a^{3n+1} / n \geq 0$$

$$(aaa)^* a \cup a(aaa)^*$$

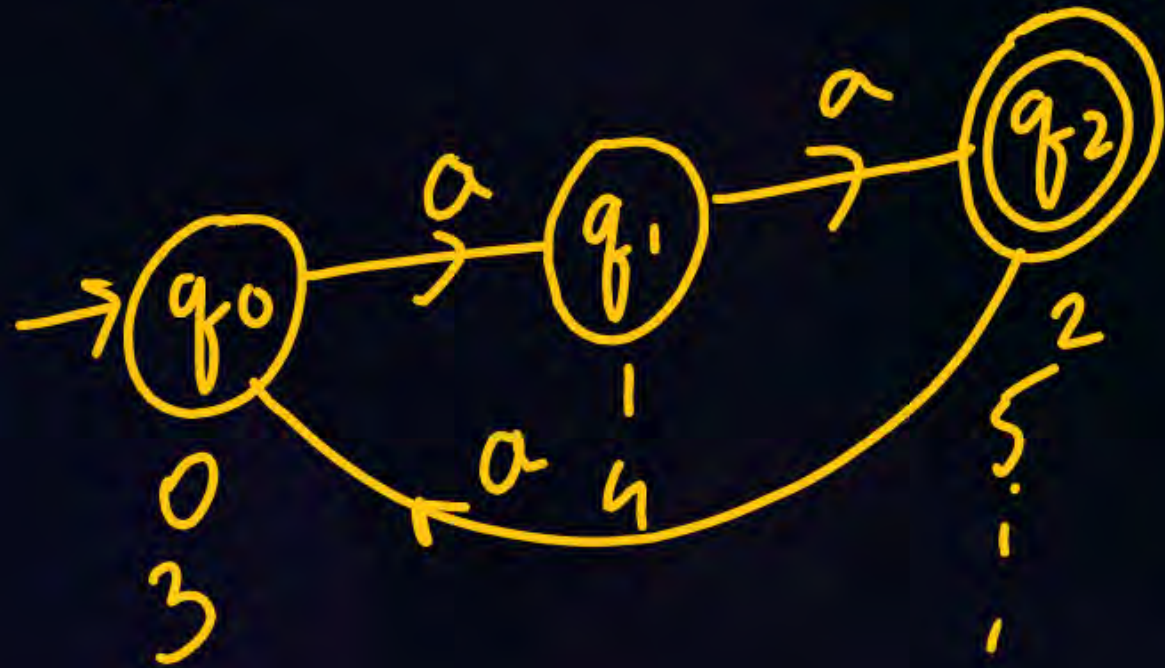
$\equiv$

Both min NFA and min DFA



$\Sigma = a$

$L = \text{no of } a\text{'s} \equiv 2 \pmod 3$

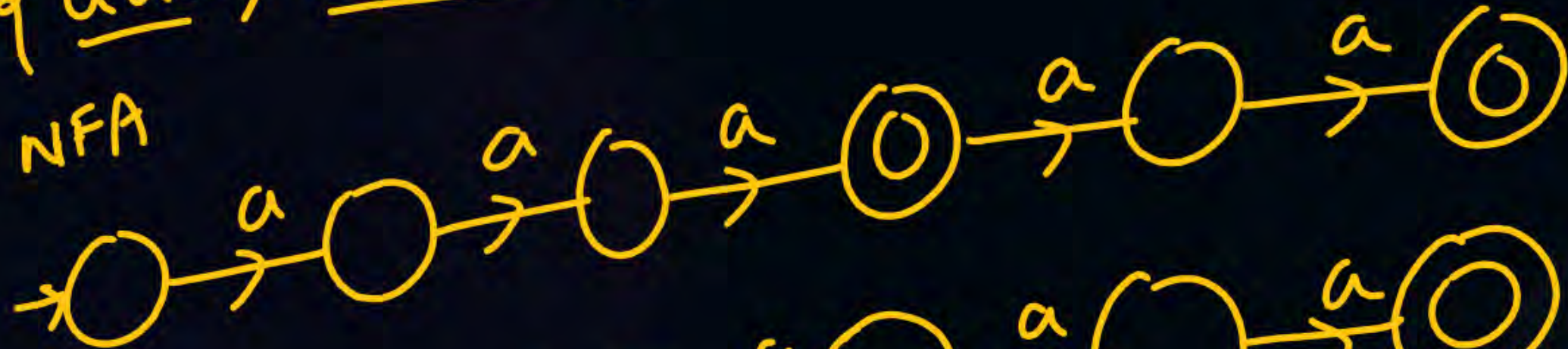


✓ $(aaa)(aaa)^*$
 $a(aaa)^*a$
 $(aaa)^*aa$

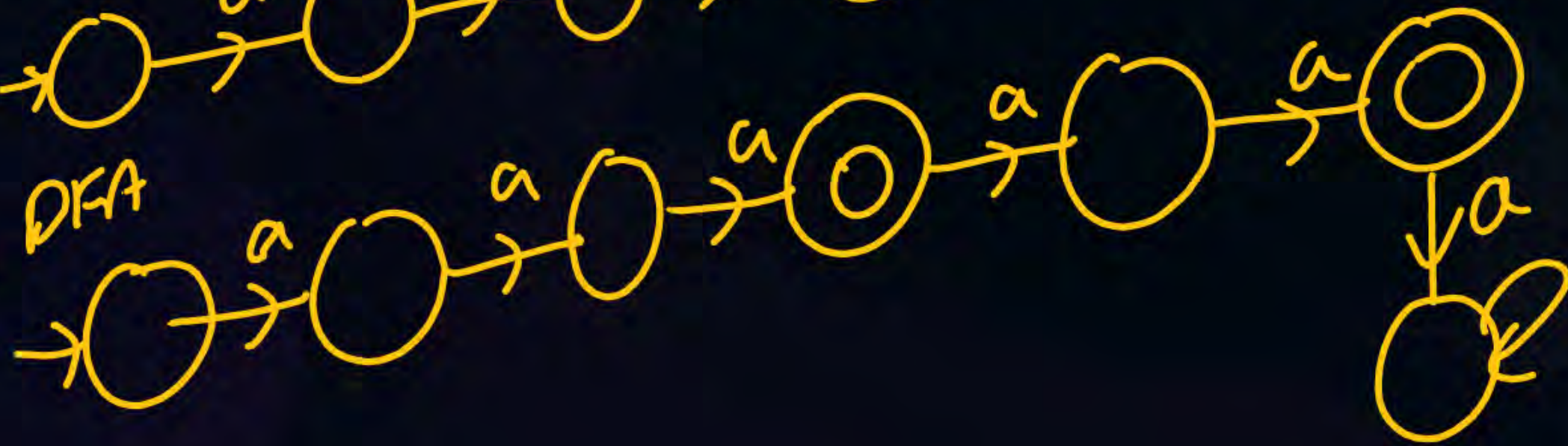
$(a^n / n=5 \Delta n=3)$

$L = \{ \underline{aaa}, \underline{aaaaa} \}$

min NFA



min DFA



$$\Sigma = \{a\}$$

$$L = (\underline{aa} + \underline{aaa})^* / \underline{a^n / n = 2x + 3y, x, y \geq 0}$$

$$L = \{\epsilon, aa, aaa, \underline{aaaa}, \underline{aaaaa}, \underline{aaaaaa}, \dots\}$$

$$L = \Sigma^* - \{a\}$$



For a given language, there may be many min NFA. But min DFA is always unique. All min NFA's will have same no of states.

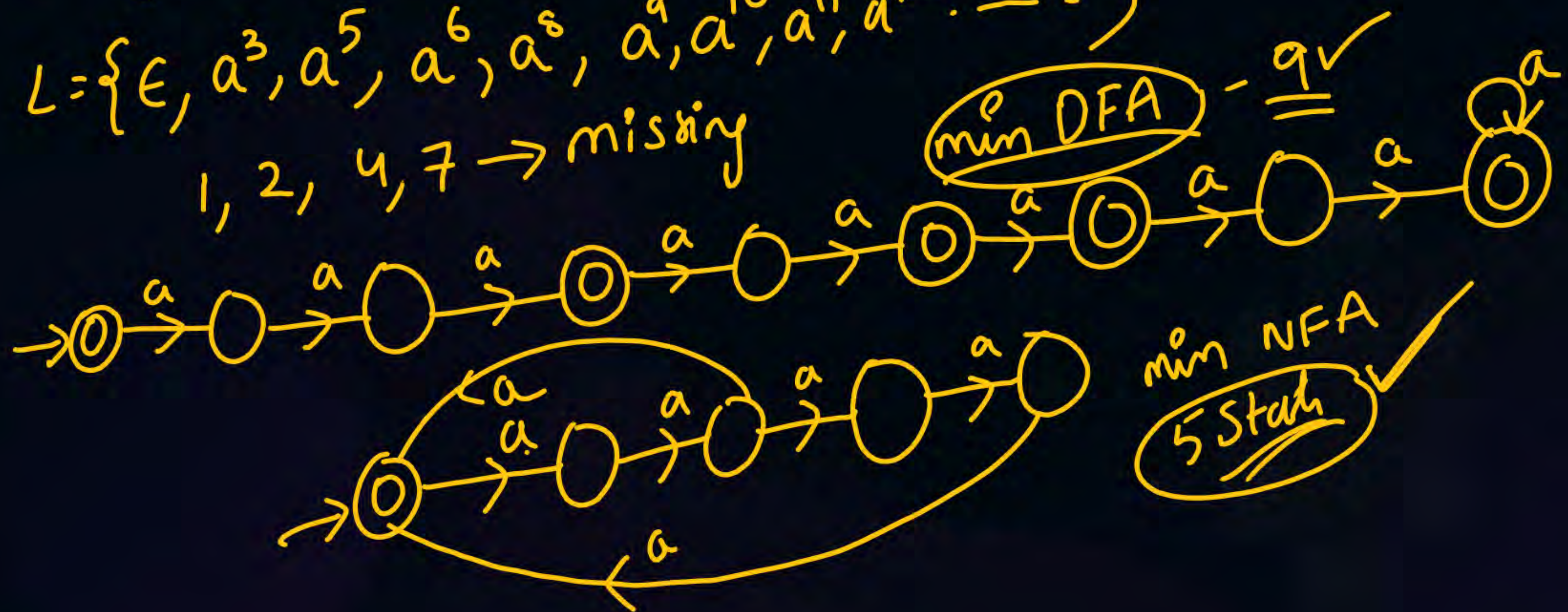
Gate:-

$$\Sigma = \{a\}$$

$$L = (\underline{aaa} + \underline{aaaaa})^* / a^n / n = 3x + 5y / x, y \geq 0$$

$$L = \{\epsilon, a^3, a^5, a^6, a^8, a^9, a^{10}, a^{11}, a^{12}, \dots\}$$

1, 2, 4, 7 $\rightarrow$ missing



min NFA
5 states



THANK - YOU